3.1.6 Rails

A PSU generates different voltages. All cables carrying the same voltage constitute a rail. In a PSU, the cables are colour-coded: the Red cable carries +5 V, the Yellow cable carries +12 V, the black cable is Ground, and so on.

A modular power supply avoids wire clutter

3.1.8 SMPS

Switching-Mode (or Switched-Mode) Power Supply refers to that class of devices that use a switching transistor (along with other components) to convert AC current into DC current of different voltages. The switching transistor is used since the mode of operation—constantly switching off and on, effectively "chopping up" the input power—is the most power-efficient. Using the same mechanism as in adapters for single devices would result in significant loss of energy as heat during the conversion process.

3.1.9 Standby Power

In ATX power supplies, the user cannot directly power up the SMPS. The SMPS is connected to the motherboard, which controls it. Standby power is the minimal power (at 5 volts) that is used to keep the motherboard active even when shut down by the user. This allows the system to be powered up manually by the user by depressing the power button on the cabinet (which actually passes a signal to the motherboard, not to the power supply), or automatically by power-up events like sending a signal over the network (Wake on LAN) or by setting a wake-up time in the motherboard BIOS.

3.1.10 Wattage

This refers to the maximum power in watts that the PSU can provide. This is a peak value; the effective power will be less than this. A PSU that provides at least 20 per cent more power than what is needed by the system is recommended.